Summary

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Definition
If function f is continuous on interval $[a, b]$ then

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{k=1}^{n} f(x_k^*)\Delta x$$

Note 1: When $f(x) \geq 0$, on the interval $[a, b]$, the definite integral, $\int_a^b f(x)\,dx$, gives
the **area** of the region between the graph of f and the interval $[a, b]$ on the $x-$axis.

Note 2: When $f(x) < 0$, on some interval in $[a, b]$, then the definite integral, $\int_a^b f(x)\,dx$,
gives the **net area** of the region between the graph of f and the interval $[a, b]$ on the $x-$axis.

Properties of Definite Integrals

(a) $\int_a^a f(x)\,dx = 0$

(b) $\int_a^c f(x)\,dx = \int_a^b f(x)\,dx + \int_b^c f(x)\,dx$

(c) $\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx$

(d) $\int_a^b k \cdot f(x)\,dx = k \cdot \int_a^b f(x)\,dx$

(e) $\int_a^b (f(x) + g(x))\,dx = \int_a^b f(x)\,dx + \int_a^b g(x)\,dx$; $\int_a^b (f(x) - g(x))\,dx = \int_a^b f(x)\,dx - \int_a^b g(x)\,dx$

Note 3: If the function f is **odd**, i.e, if $f(-x) = -f(x)$, for all x in $[-a, a]$, then

$$\int_{-a}^{a} f(x)\, dx = 0$$

Note 4: If the function f is **even**, i.e, if $f(-x) = f(x)$, for all x in $[-a, a]$, then

$$\int_{-a}^{a} f(x)\, dx = 2 \cdot \int_{0}^{a} f(x)\, dx$$